

# Corolario de maximización acotada

**Corolario 1 (Maximización acotada).** *Sea  $f: \mathbb{N}^{k+1} \rightarrow \mathbb{N}$  una función recursiva primitiva. Entonces,  $Mf: (n, x) \mapsto \max(\{k \leq n : f(k, x) = 0\} \cup \{0\})$  es una función recursiva primitiva.*

***Demostración:*** Podemos expresar  $Mf$  en términos de la minimización acotada de  $f$ :

$$\max\{k \leq n : f(k, x) = 0\} = n - \min\{k \leq n : f(n - k, x) = 0\}.$$

Entonces, por el lema de minimización acotada,  $Mf$  es recursiva primitiva. ■

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